aircraft over the earth's surface projected forward from a predetermined initial position known at the time of takeoff. No external navigation aids except true airspeed inputs from the Central Air Data Computer (CADC) were required by the INS.

The Continental Boeing 747 utilized three (3) separate Litton LTN-51 Inertial Navigation Systems. Synchro and binary signals selected for recording were provided by the No. 3 INS. Figure 2 illustrates the parameters that were received from these outputs. The No. 3 INS Control Display Unit (CDU) was located on the aft central pedestal. From the jump seat, the meteorological observer could operate the data selector switch and manually record the various INS parameters that were provided continuously in a numerical display. This enabled the observer to note real time data associated with a turbulence event.

Central Air Data Computer

The Central Air Data Computer (CADC) supplied computed air data from information received from dynamic (pitot) pressure, ambient (static) pressure, and total air temperature which were sensed external to the aircraft. With the availability of dual systems on the B-747, selected outputs from the No. 2 Bendix CADC were connected to the FDAU for signal conditioning prior to recording. The same data were transmitted to the aircraft's flight control, inertial navigation, and air data instrumentation systems. The computed air data functions that were recorded included altitude (coarse and fine), true airspeed (TAS), computed airspeed (CAS), Mach number, and static air temperature (SAT).

Greenwich Mean Time Clock

A Hayden Greenwich Mean Time (GMT) clock with an output for recording time was installed in the captain's panel in place of the original Boeing 747 clock. The primary purpose of the clock was to provide a GMT digital readout for the meteorological observer. In addition, the GMT clock output signal was transmitted to the FDAU and ARIS II for conditioning and subsequent recording as a backup for the ARIS II clock. Before each flight, the GMT clock was set with the correct time obtained by radio from the National Bureau of Standards, Station WWV. The ARIS II electronic clock was then started and set with the identical time.